Lunar Smart Adaptive Habitat Design (NASA HUNCH) (with our notes)

The NASA HUNCH project page, with notes under each page about what we built for that requirement, plus screenshots from both games.

The physical parts (the 2 ft by 2 ft model, foam board and sensor electronics) are made outside the games, and we marked those.



Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring for Crew Adaptive Positive Mental Health

2025-26

BIOMEDICAL
And / Or
Software Projects or Both

Done | Lunar Smart Adaptive Habitat Design

- We made two working Unity/VR games, plus the physical model.

Screenshots:



Realistic game



Done | Biomedical and software

- We did both. Software: the AI system and the two games. Biomedical: made-up wearable data for each crew member (heart rate, HRV, sleep, skin response, cortisol).

Screenshots:



Realistic game



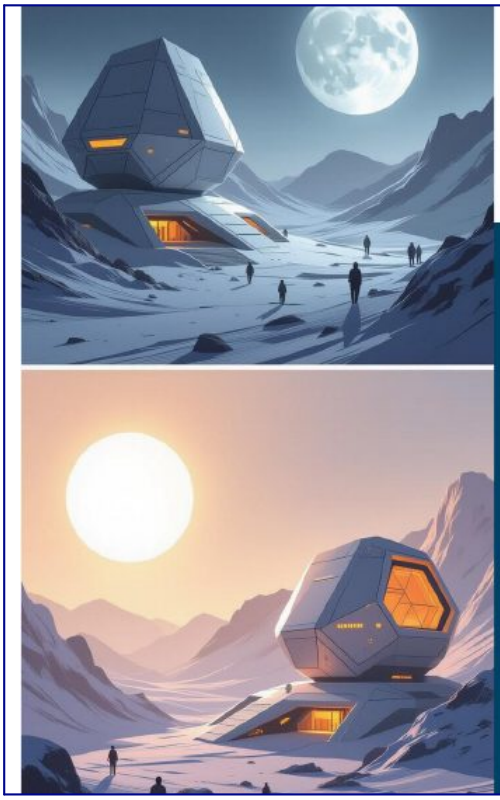
Done | Project overview

- The AI uses psychology research to look after the crew's mental health. It's connected to the habitat systems (life support, power, environment) in the same game.

Screenshots:



Realistic game

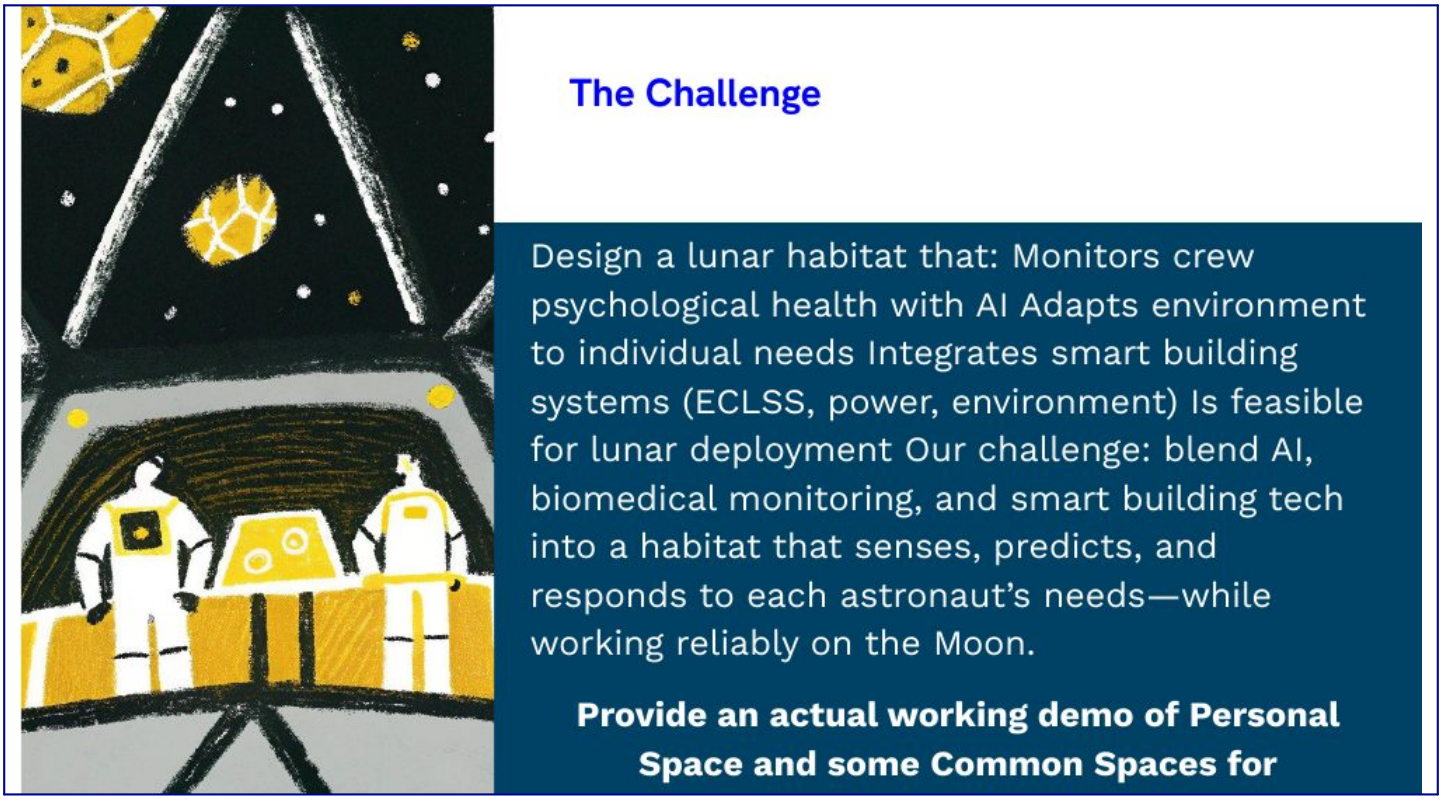


The Problem

Long lunar missions = isolation, stress, and risk to mental health Current habitats: focus on physical safety, not psychological needs Need: Adaptive, supportive environments for crew well-being Astronauts face unique psychological challenges: isolation, confinement, and stress. Traditional habitats protect physically, but we must also address mental health for mission success.

Done | The problem

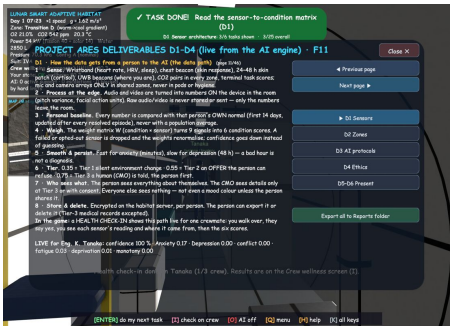
- Every room and every AI action is aimed at one of the six risks: isolation, stress, being stuck inside and the rest.



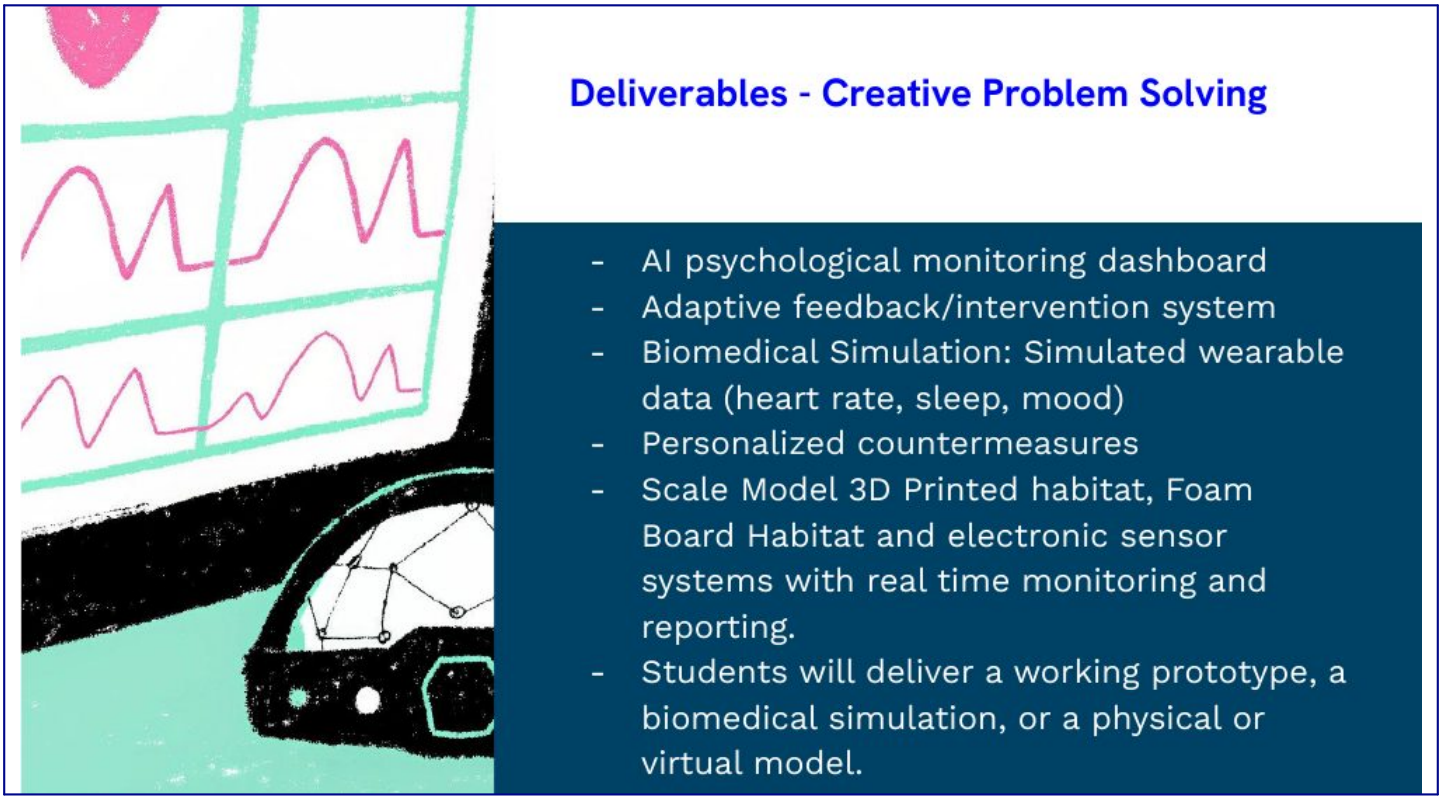
Done | The challenge

- Monitor: the live AI dashboard. Adapt: the AI changes things for each person. Integrate: life support, power and alarms are in the same game. Realistic: every sensor has a weight and power cost.

Screenshots:



Realistic game



Done | Deliverables

- AI mental health dashboard: yes (press F1).
- System that adapts and helps: yes (three tiers, offers and crew tasks).
- Biomedical simulation with fake wearable data: yes. The health check-in shows heart rate, HRV, sleep, skin response and cortisol.
- Personal help: yes. "Help" picks something that fits what that person is struggling with.
- Demo of private and shared spaces: yes (the sleep dome and the galley).

Still need to do: The scale model, foam board and sensor electronics are physical, so they're made outside the game (Farhan's model).

Screenshots:



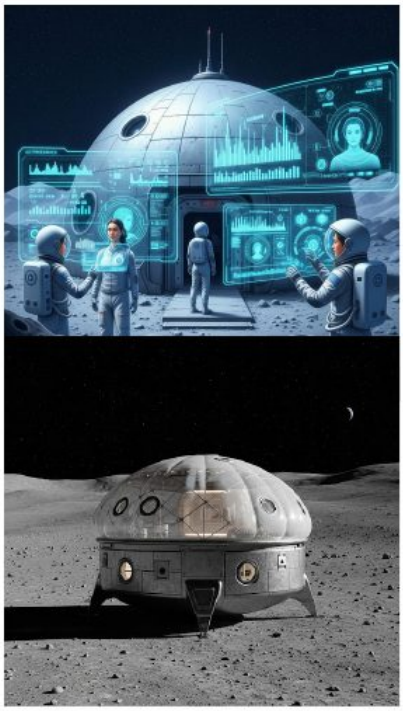
Realistic game



Realistic game



Realistic game, daily wellness tasks



So What would I do First?

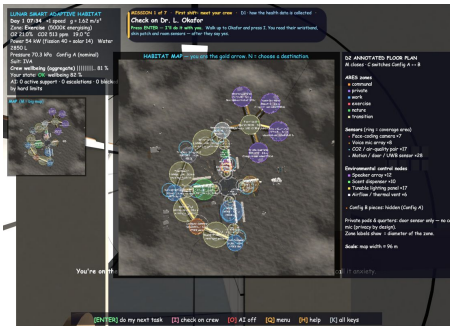
- Research NASA Pill Habitat Design for the moon

Feature	Details (Based on NASA Artemis/Surface Habitat Concepts)
Crew Capacity	2 astronauts (standard), expandable to 4 (peak)
Uncrewed Duration	Up to 6–24 months of autonomous operation
Structure	Multi-level: rigid metallic base + inflatable upper deck
Internal Layout	Sleeping, food prep, hygiene, work, storage
Life Support	Advanced ECLSS, waste management, radiation shielding
Power	Solar arrays + nuclear fission backup
Location	Lunar south pole (e.g., Shackleton Crater)
Autonomy	Remote monitoring, automated maintenance, and mode switching

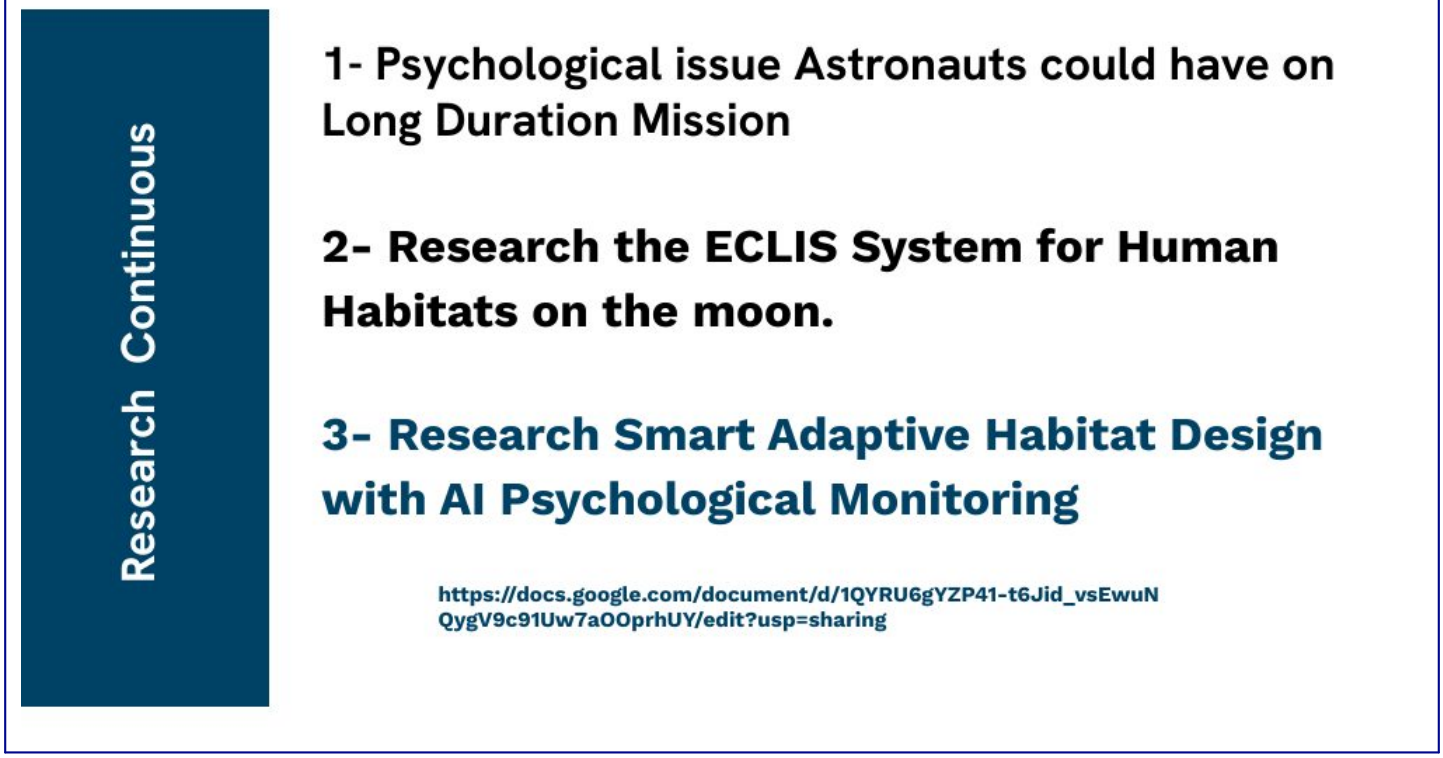
Done | NASA pill habitat research

- 4 crew fits the 2 to 4 limit. Our base has multiple levels, life support, solar plus nuclear power, and sits at the south pole.

Screenshots:



Realistic game



Done | Research

- Our 64 page research doc is where the numbers in the games came from.





Make a Plan and ....

Sketch out a few designs for the outside

Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print

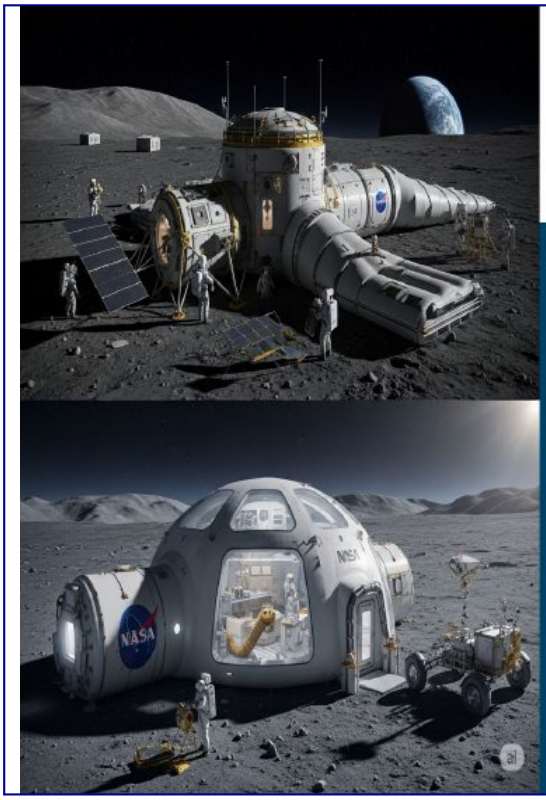
Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

Done | Make a plan

- In Unity you can walk into every room. The physical scale model is separate.



Make a Plan and ....

Sketch out a few designs for the outside

Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print
- Other

Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

Done | Make a plan (continued)

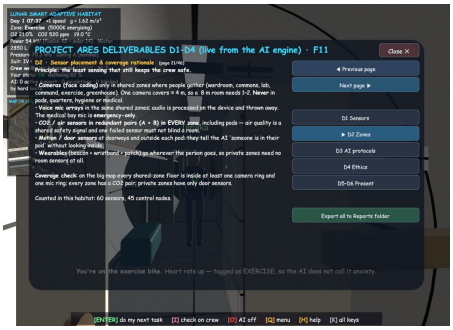
- Same as the last page.



Done | Smart, adaptive mental health monitoring

- Voice and face analysis only happen in shared rooms, and it's all processed in the room. Location sensors show who's spending time together and who's alone. We planned it room by room.

Screenshots:



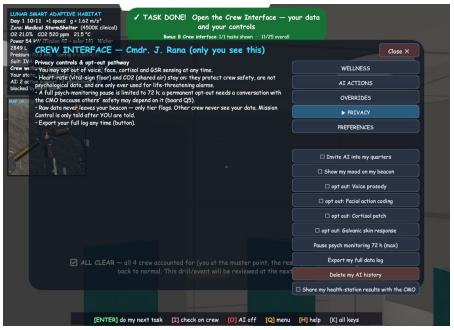
Realistic game



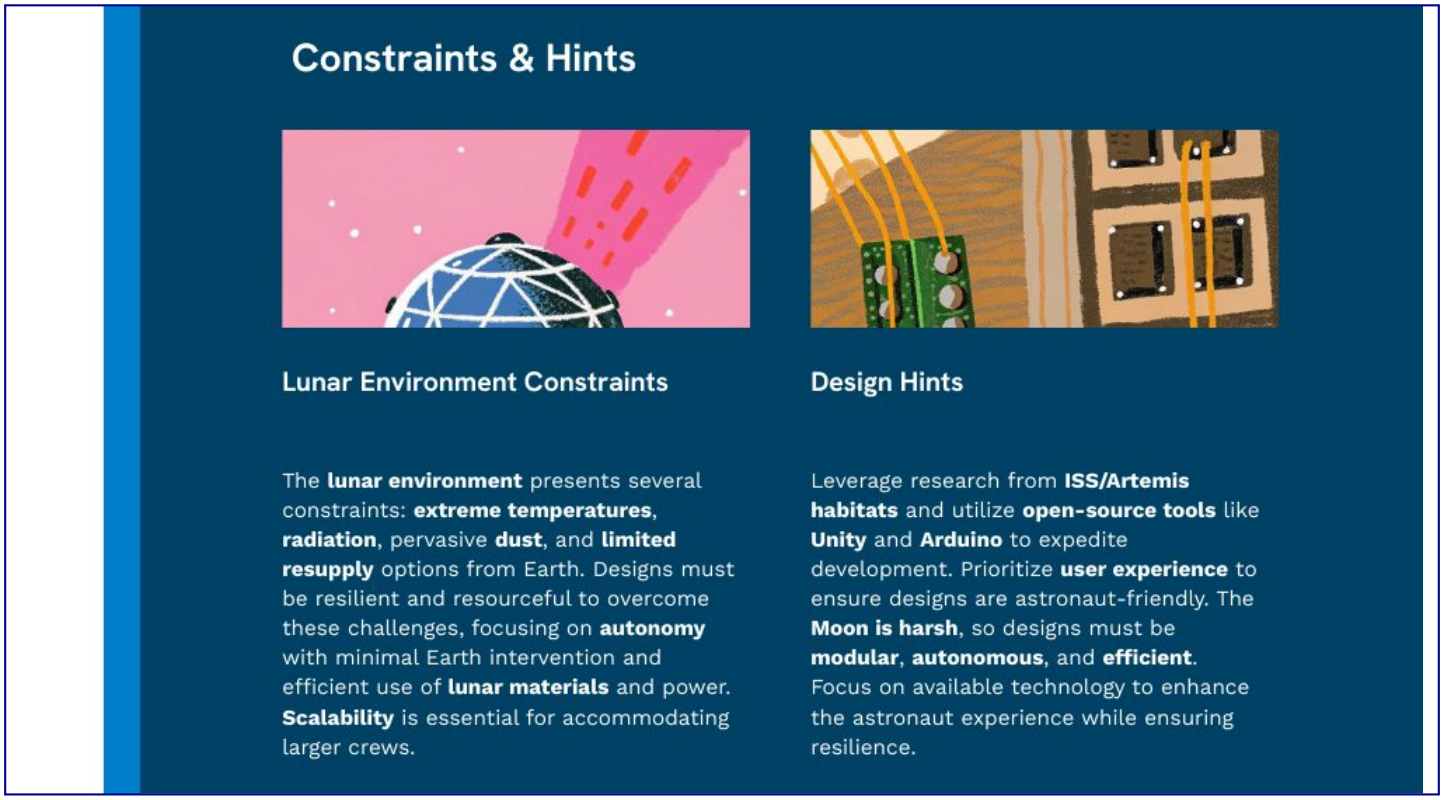
Done | Key things to consider

- Privacy is respected: no sensors in pods, plus opt-out, export and delete. VR works on the Meta Quest, and there's adaptive lighting and social suggestions.

Screenshots:



Realistic game



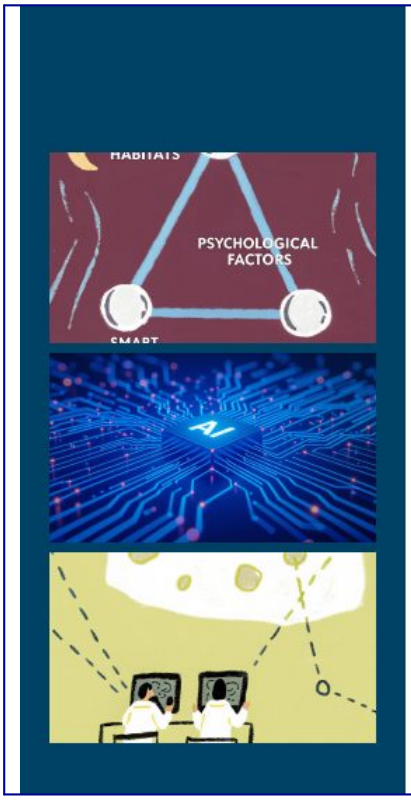
Done | Constraints and hints

- We used Unity like the page suggests. We also made it easier for astronauts to use: one key (ENTER or G) does your next task, the menu is simple, and the key bar is short.

Screenshots:



Realistic game, simple menu



Example Project Flow

Research

Conduct thorough **research** on **psychological factors**, **lunar habitats**, and **smart systems** to lay a solid foundation for the project.

More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

Design

Create a detailed **design** by sketching the **layout**, defining **AI parameters**, and planning **integration** to guide the development process.

Prototype, Test, Present

Develop a **model/VR** prototype, **develop software**, **simulate scenarios**, **refine interventions**, and then **present** a **demo**, **walkthrough**, and **report**.

Done | Example project flow

- Prototype: two games. Testing: we played through every mission and button again and again. Presenting: the presentation order is built into the game.

Screenshots:



Realistic game



Conclusion

Students innovate for real NASA needs Combining, biomedical, software and engineering skills Supports future lunar explorers' mental and physical health This project is a chance to make a real impact on the future of space exploration. By supporting astronaut well-being, you're helping pave the way for humanity's next giant leap.

Done | Conclusion

- Biomedical and software come together in one demo.



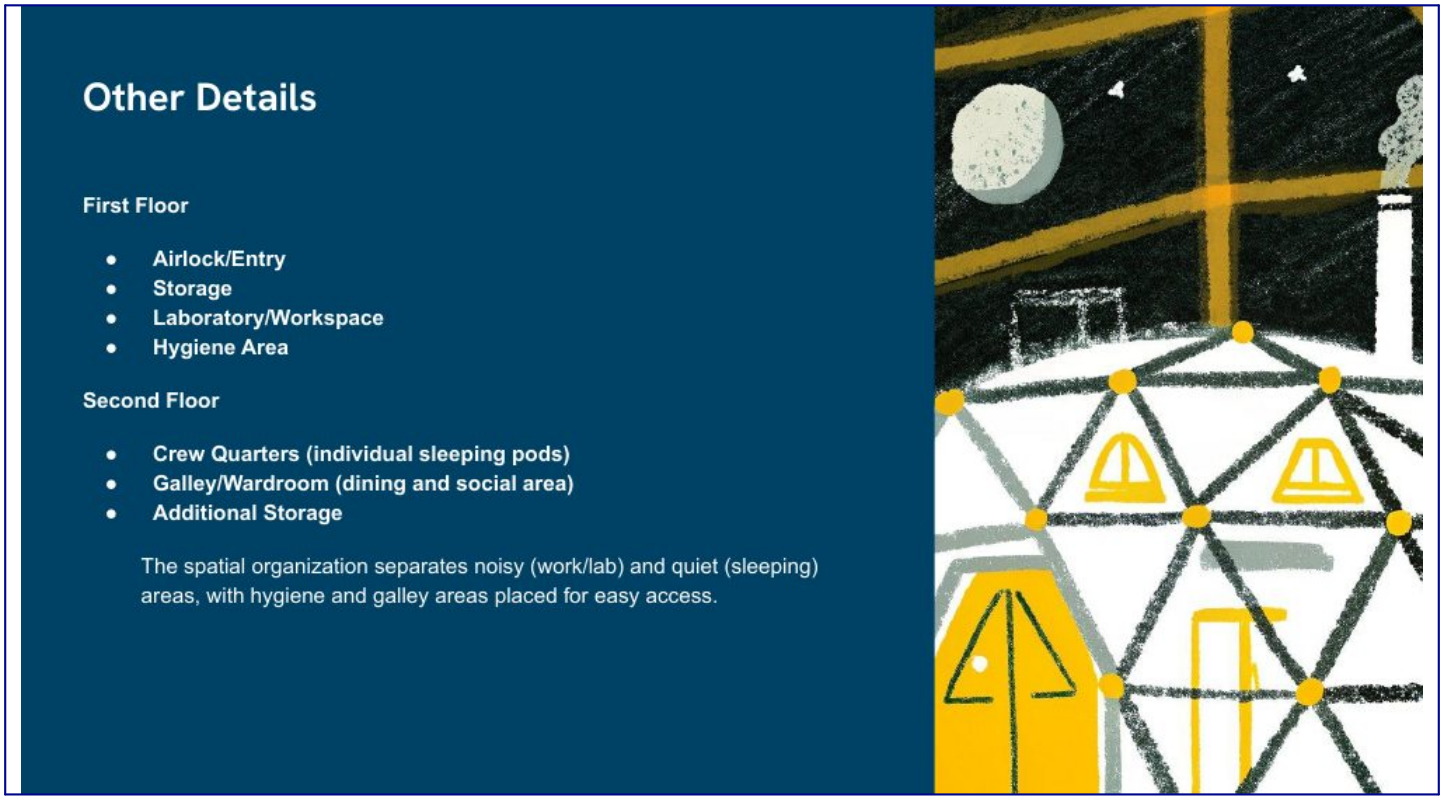
Done | Questions

- We practiced board questions Q1 to Q6 live.

Screenshots:



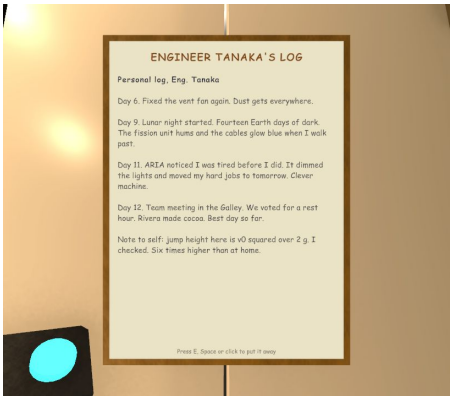
Realistic game



Done | Floor layout: quiet vs noisy

- Sleeping areas are away from the work and lab areas, and the bathroom and kitchen are easy to get to.

Screenshots:



Sci-fi game, crew quarters (center dorm kept)

Other Details - Room/Space - Recommended Dimensions (approximate)

Crew Quarters

2.1 m x 2.1 m x 2.4 m (7' x 7' x 8')

Galley/Wardroom

3.0 m x 3.0 m x 2.4 m (10' x 10' x 8')

Hygiene Area

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Laboratory/Workspace

4.0 m x 3.0 m x 2.4 m (13' x 10' x 8')

Storage

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Airlock

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Include: EVA Suit and Vehicle Port



Mostly done | Room sizes

- Our rooms are close to these sizes: about 2.1 m sleep pods, a 3 m kitchen, a 4 by 3 m lab, and an airlock with a suit port.

Still need to do: We should put the exact measurements on the printed floor plans.

Research

Key for this project is to start to Research in:

- **Psychological Support**: VR/AR, adaptive lighting, social prompts
- **AI Monitoring**: Real-time, privacy-respecting, adaptive, Psychological and Physical
- **Smart Systems**: ECLSS, environmental, and power controls
- **Structure**: Modular, expandable, radiation-shielded



Done | Research areas

- All four areas are in the games: VR nature and lighting for mental health, a privacy-friendly AI, life support and power controls, and a modular base.